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GRAPHIC COUNTEREXAMPLES TO KAHN'S BASIS CONJECTURE IN EVERY RANK AT LEAST FOUR

GPT-5.6 SOL PRO

ABSTRACT. For every integer $n \geq 4$, we construct an $n \times n$ array of bases of a rank- n graphic matroid for which no choice of one element from each cell makes every row and every column a basis. The matroid is the cycle matroid of the wheel with rim length n , and the array has a fixed block form involving four spanning trees. An exact criterion for this block form reduces the selection problem to two boundary matchings arising from single-element basis exchanges, together with a common completion at the bottom-right cell. Alternating edge decompositions of even and odd wheels make these completion conditions incompatible. A reduced signed incidence matrix realizes each counterexample over every field.

We also prove the selection property for partition matroids, for matroids of rank at most two, and for rank-three matroids whose nonloop restriction is disconnected. An exhaustive computation establishes the property for the Fano matroid and hence for every rank-three binary matroid. It follows that rank four is minimal among binary, regular, and graphic matroids. Further exhaustive searches give restricted minimality results for the smallest wheel examples.

STATEMENT ON HUMAN USE

The results of this paper were obtained after multiple prompts from Ho Boon Suan. The correctness of the constructions was verified by the author. The paper was written by the author (humans were used for minor edits to improve grammar and style).

1. INTRODUCTION

All matroids in this paper are finite. Let M be a matroid of rank $n \geq 1$, and assign a basis B_{ij} of M to each cell (i, j) of an $n \times n$ array. A *selection* is a family

$$x_{ij} \in B_{ij} \quad (1 \leq i, j \leq n).$$

It is *successful* if, for every i and j , both

$$\{x_{i1}, \dots, x_{in}\} \quad \text{and} \quad \{x_{1j}, \dots, x_{nj}\}$$

are bases of M .

Huang and Rota recorded the corresponding vector-space conjecture, attributed to Kahn, over arbitrary infinite fields [4, Conjecture 6]. Bucić, Kwan, Pokrovskiy, and Sudakov subsequently formulated the matroidal version and proved an asymptotic partial result [1, 2].

Conjecture 1.1 (Kahn's basis conjecture, matroidal form). *Every $n \times n$ array of bases of a rank- n matroid has a successful selection.*

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Rota's basis conjecture is the column-constant special case $B_{ij} = B_j$. In that setting, a successful selection uses every element of B_j once in column j , and its rows are transversal bases of $B_1, \dots, B_n$.

The main result gives counterexamples to Conjecture 1.1 in every rank at least four.

Theorem 1.2. *For every integer $n \geq 4$, the cycle matroid of the wheel with rim length n admits an $n \times n$ array of bases with no successful selection. The array uses four distinct spanning trees, and the wheel has $2n$ edges.*

Corollary 1.3. *For every field $\mathbb{F}$ and every integer $n \geq 4$, there is an $n \times n$ array of bases of $\mathbb{F}^n$ with no successful selection. For each n , a single integral reduced incidence matrix realizes the construction over every field.*

The construction has the same block form in every rank: the top-left $(n-1) \times (n-1)$ block is filled with a basis X , the remaining cells in the first $n-1$ rows use a basis Y , the remaining cells in the first $n-1$ columns use a basis Z , and the bottom-right cell uses a basis W . Proposition 2.2 characterizes successful selections in such an array. Its forward direction extracts a common multiset of omissions from the X -block and turns the row and column conditions into two systems of single-element exchanges. Its converse uses Smetaniuk's theorem on completing partial Latin squares to reconstruct the interior of the array.

For a wheel, each allowable exchange can be read from a fundamental cut of a spanning tree. In even rank, alternating spokes and rim edges give a cyclic pattern with no defect. The boundary matchings then determine all but one omitted edge, and the two possible boundary completions require incompatible corner elements. In odd rank, one spoke and one rim edge repair the alternating pattern. The possible omission multisets remain sufficiently rigid to admit an explicit case analysis, again with incompatible corner requirements.

The following results locate the first possible rank of a counterexample more precisely. For a matroid M , let $L(M)$ denote its set of loops.

Theorem 1.4. *Every array of bases of a matroid of rank at most two has a successful selection. The same conclusion holds for a rank-three matroid M whenever $M \setminus L(M)$ is disconnected.*

Theorem 1.5 (computer-assisted). *Every 3×3 array of bases of a rank-three binary matroid has a successful selection.*

Corollary 1.6. *Among binary matroids, among regular matroids, and among graphic matroids, the least rank of a counterexample is four.*

Section 2 proves the block criterion. Sections 3 and 4 give the even- and odd-rank wheel constructions, respectively, and Section 5 records explicit coordinates for the rank-four example. Section 6 proves the positive results for low-rank and partition matroids. The exhaustive computations and their precise scope are described in Section 7. Thus rank three is the only rank not settled for arbitrary matroids.

2. AN EXACT CRITERION FOR THE BLOCK ARRAY

Let $n \geq 2$, and let X, Y, Z, W be bases of a rank- n matroid M . Define

$$\mathcal{A}_n(X, Y, Z, W) = \begin{pmatrix} X & \cdots & X & Y \\ \vdots & & \vdots & \vdots \\ X & \cdots & X & Y \\ Z & \cdots & Z & W \end{pmatrix}, \quad (1)$$

where the top-left block has size $(n-1) \times (n-1)$. For $x \in X$ and an element y of the ground set, write

$$X - x + y = (X \setminus \{x\}) \cup \{y\}.$$

Let Ω be a multiset of $n-1$ elements of X . A Y -matching missing $e \in Y$ is a bijection from the occurrences of Ω to $Y \setminus \{e\}$ such that an occurrence of x may be matched to y only if $X - x + y$ is a basis. A Z -matching missing $f \in Z$ is defined analogously.

The following filling lemma is the only ingredient needed to reconstruct the top-left block from its row and column omissions.

Lemma 2.1. *Let S be a set of size n , and let*

$$\rho_1, \dots, \rho_{n-1}, \quad \kappa_1, \dots, \kappa_{n-1}$$

be elements of S such that the two lists agree as multisets. There is an $(n-1) \times (n-1)$ array with entries in S whose i th row contains each element of $S \setminus \{\rho_i\}$ exactly once and whose j th column contains each element of $S \setminus \{\kappa_j\}$ exactly once.

Proof. Set $m = n-1$, and choose an element $q \in S$ that occurs in the common omission multiset. For each $x \in S \setminus \{q\}$, pair bijectively the rows omitting x with the columns omitting x , and place the symbol x in the corresponding cells. These prescribed entries form a partial Latin square of order m on the symbol set $S \setminus \{q\}$: no symbol is repeated in a row or column. The number of prescribed cells is

$$m - |\{i : \rho_i = q\}| \leq m - 1.$$

Smetaniuk's theorem therefore completes the partial Latin square [6]. In the completed square, replace every prescribed occurrence of x by q . A row originally omitting $x \neq q$ loses its unique occurrence of x and gains one occurrence of q , whereas a row omitting q is unchanged. The same argument applies to the columns, and the resulting array has the required omission lists. $\square$

Proposition 2.2 (block criterion). *The block array $\mathcal{A}_n(X, Y, Z, W)$ has a successful selection if and only if there are a multiset Ω of $n-1$ elements of X and elements $e \in Y$, $f \in Z$, and $w \in W$ such that*

- (i) Ω has a Y -matching missing e ;
- (ii) Ω has a Z -matching missing f ;
- (iii) both $Y - e + w$ and $Z - f + w$ are bases.

Proof. Suppose first that a successful selection is given. In each of the first $n-1$ rows, the selections from the X -block are $n-1$ distinct elements of X , hence equal $X \setminus \{\rho_i\}$ for a unique $\rho_i \in X$. Define $\kappa_j \in X$ analogously from the first $n-1$ columns. If N_x is the number

of selected occurrences of $x \in X$ in the top-left block, then counting by rows and by columns gives

$$N_x = (n - 1) - |\{i : \rho_i = x\}| = (n - 1) - |\{j : \kappa_j = x\}|.$$

Thus the row and column omission lists agree as multisets; denote their common multiset by Ω .

The selected elements in the first $n - 1$ cells of the last column are distinct elements of Y , so they form $Y \setminus \{e\}$ for some $e \in Y$. The basis condition in row i says precisely that the selected boundary element may be matched to the omission ρ_i , giving a Y -matching of Ω missing e . The last row similarly gives a Z -matching missing some $f \in Z$. If w is selected in the bottom-right cell, then the last column and last row are the bases $Y - e + w$ and $Z - f + w$.

Conversely, assume that Ω, e, f, w satisfy the three conditions. Order the occurrences of Ω as

$$\rho_1, \dots, \rho_{n-1} \quad \text{and} \quad \kappa_1, \dots, \kappa_{n-1}.$$

Use the Y -matching to assign the elements of $Y \setminus \{e\}$ to the first $n - 1$ rows, and use the Z -matching to assign the elements of $Z \setminus \{f\}$ to the first $n - 1$ columns. Lemma 2.1 fills the top-left block with row omissions ρ_i and column omissions κ_j . The first $n - 1$ rows and columns are then the prescribed single-element exchanges of X and hence are bases. Selecting w in the bottom-right cell completes the last row and column by condition (iii). $\square$

Only the forward implication is needed to rule out successful selections in the wheel constructions. The converse makes the criterion exact and is used in the exhaustive searches of Section 7.

3. EVEN RANKS

Let $n = 2m$ with $m \geq 2$, and write $\mathbb{Z}_t = \mathbb{Z}/t\mathbb{Z}$. Let G_n be the wheel with hub h and rim vertices indexed by $\mathbb{Z}_{2m}$. Denote its spokes and rim edges by

$$s_j = \{h, j\}, \quad r_j = \{j, j + 1\},$$

where rim-vertex indices are read modulo $2m$. For $i \in \mathbb{Z}_m$, set

$$a_i = r_{2i+1}, \quad b_i = s_{2i}, \quad c_i = s_{2i+1}, \quad d_i = r_{2i}. \quad (2)$$

Let

$$\begin{aligned} A &= \{a_i : i \in \mathbb{Z}_m\}, & B &= \{b_i : i \in \mathbb{Z}_m\}, \\ C &= \{c_i : i \in \mathbb{Z}_m\}, & D &= \{d_i : i \in \mathbb{Z}_m\}, \end{aligned}$$

and define

$$X = A \cup B, \quad Y = A \cup C, \quad Z = B \cup D, \quad W = C \cup D. \quad (3)$$

Subscripts on a_i, b_i, c_i, d_i are henceforth read modulo m .

Lemma 3.1. *The sets X, Y, Z, W in (3) are spanning trees of G_n .*

Proof. Each set has $2m$ edges, one fewer than the number of vertices of G_n . In X , every even rim vertex is joined to h by its spoke in B , and the odd vertex $2i + 1$ is attached as a leaf to $2i + 2$ by a_i . In Z , the same even spokes are used, while d_i attaches $2i + 1$ to $2i$. In Y and W , every odd rim vertex is joined to h by its spoke in C ; the even vertex $2i$ is attached to

$2i - 1$ by a_{i-1} in Y and to $2i + 1$ by d_i in W . Each set is therefore connected and hence is a spanning tree. $\square$

For $x \in X$, define

$$\begin{aligned} E_Y(x) &= \{y \in Y : X - x + y \text{ is a spanning tree}\}, \\ E_Z(x) &= \{z \in Z : X - x + z \text{ is a spanning tree}\}. \end{aligned}$$

For $e \in Y$ and $f \in Z$, define

$$\begin{aligned} E_W^Y(e) &= \{w \in W : Y - e + w \text{ is a spanning tree}\}, \\ E_W^Z(f) &= \{w \in W : Z - f + w \text{ is a spanning tree}\}. \end{aligned}$$

These sets are read directly from fundamental cuts: after deleting an edge from a tree, an edge of the target tree is admissible exactly when it joins the two resulting components.

Lemma 3.2. *For every $i \in \mathbb{Z}_m$,*

$$\begin{array}{c|c|c} x & E_Y(x) & E_Z(x) \\ \hline a_i & \{a_i, c_i\} & \{d_i\} \\ b_i & \{c_{i-1}\} & \{b_i, d_{i-1}, d_i\} \end{array} \quad (4)$$

and

$$\begin{array}{c|c|c|c} e & E_W^Y(e) & f & E_W^Z(f) \\ \hline a_i & \{d_{i+1}\} & b_i & \{c_i\} \\ c_i & \{c_i, d_i, d_{i+1}\} & d_i & \{d_i, c_i\}. \end{array} \quad (5)$$

Proof. For the six types of deletion that occur, the component not containing h is as follows:

$$\begin{array}{c|c} X - a_i & \{2i + 1\} \\ Y - a_i & \{2i + 2\} \\ Z - b_i & \{2i, 2i + 1\} \end{array} \quad \begin{array}{c|c} X - b_i & \{2i - 1, 2i\} \\ Y - c_i & \{2i + 1, 2i + 2\} \\ Z - d_i & \{2i + 1\}. \end{array}$$

The edges in the relevant target tree that cross each displayed cut are exactly those listed in (4) and (5). $\square$

Proposition 3.3. *For every even $n \geq 4$, the block array $\mathcal{A}_n(X, Y, Z, W)$ formed from (3) has no successful selection.*

Proof. Assume that a successful selection exists, and let Ω be the common omission multiset given by Proposition 2.2. It has size $2m - 1$. On the Y boundary, every occurrence of b_i must be matched to c_{i-1} , so b_i occurs at most once in Ω . On the Z boundary, every occurrence of a_i must be matched to d_i , so a_i also occurs at most once. Since $X = A \cup B$ has $2m$ elements,

$$\Omega = X \setminus \{q\}$$

for some $q \in X$.

First suppose that $q = b_k$. On the Z boundary, the elements a_i force all of D ; each remaining b_i , with $i \neq k$, is then forced to match itself. Thus the image of the Z -matching is $Z \setminus \{b_k\}$, and (5) forces the corner element to be c_k .

On the Y boundary, the elements b_i with $i \neq k$ force all of C except c_{k-1} . The image of the Y -matching is therefore either $Y \setminus \{c_{k-1}\}$ or $Y \setminus \{a_{k-1}\}$. By (5), the corresponding sets of possible corner elements are

$$\{c_{k-1}, d_{k-1}, d_k\} \quad \text{and} \quad \{d_k\},$$

respectively. Neither contains c_k , because $m \geq 2$.

Now suppose that $q = a_k$. Every b_i forces c_{i-1} on the Y boundary, so all of C is used and the image is $Y \setminus \{a_k\}$. The corner element is therefore forced to be d_{k+1} . On the Z boundary, the elements a_i with $i \neq k$ force all of D except d_k . The image of the Z -matching must be one of

$$Z \setminus \{d_k\}, \quad Z \setminus \{b_k\}, \quad Z \setminus \{b_{k+1}\}.$$

The corresponding corner-completion sets are

$$\{d_k, c_k\}, \quad \{c_k\}, \quad \{c_{k+1}\},$$

none of which contains d_{k+1} . Both possibilities for q are impossible. $\square$

4. ODD RANKS

Let $n = 2m + 1$ with $m \geq 2$. Let G_n be the wheel with hub h , rim vertices $0, 1, \dots, 2m$, spokes s_j , and rim edges $r_j = \{j, j+1\}$, where $r_{2m} = \{2m, 0\}$. Set

$$p = s_0, \quad t = r_{2m},$$

and, for $0 \leq i < m$, define

$$a_i = r_{2i}, \quad b_i = s_{2i+2}, \quad c_i = s_{2i+1}, \quad d_i = r_{2i+1}. \quad (6)$$

Let

$$\begin{aligned} A &= \{a_i : 0 \leq i < m\}, & B &= \{b_i : 0 \leq i < m\}, \\ C &= \{c_i : 0 \leq i < m\}, & D &= \{d_i : 0 \leq i < m\}, \end{aligned}$$

and define

$$\begin{aligned} X &= \{p\} \cup A \cup B, & Y &= A \cup C \cup \{t\}, \\ Z &= B \cup D \cup \{t\}, & W &= C \cup D \cup \{t\}. \end{aligned} \quad (7)$$

The spoke p closes the alternating construction in X , while the rim edge t plays the corresponding role in Y, Z, W .

Lemma 4.1. *The sets X, Y, Z, W in (7) are spanning trees of G_n .*

Proof. Each set has $2m + 1$ edges on $2m + 2$ vertices, so it is enough to verify connectivity. In X , every even rim vertex has its spoke, and a_i attaches the odd vertex $2i + 1$ to $2i$. In Y , every odd vertex has its spoke; the edges a_i attach the even vertices $0, 2, \dots, 2m - 2$, and the path through t and a_0 attaches $2m$. In Z , the positive even vertices have the spokes in B , each d_i attaches $2i + 1$ to $2i + 2$, and t attaches 0 to $2m$. In W , the odd vertices have the spokes in C , each d_i attaches $2i + 2$ to $2i + 1$, and t attaches 0 to $2m$. Hence all four sets are connected. $\square$

Use the exchange notation E_Y, E_Z, E_W^Y, E_W^Z introduced in Section 3. In the next table, set $d_{-1} = d_{m-1}$.

Lemma 4.2. *The exchanges from X are*

x	$E_Y(x)$	$E_Z(x)$
p	$\{c_0, t\}$	$\{d_0, t\}$
a_i	$\{a_i, c_i\}$	$\{d_i\} \quad (0 \leq i < m)$
b_i	$\{c_{i+1}\}$	$\{b_i, d_i, d_{i+1}\} \quad (0 \leq i < m - 1)$
b_{m-1}	$\{t\}$	$\{b_{m-1}, d_{m-1}, t\}.$

(8)

The corner exchanges are

$$\begin{array}{c|c|c|c} e & E_W^Y(e) & f & E_W^Z(f) \\ \hline a_i & \{d_{i-1}\} & b_i & \{c_i\} \\ c_i & \{c_i, d_{i-1}, d_i\} & d_i & \{d_i, c_i\} \\ t & \{t, d_{m-1}\} & t & \{t\}. \end{array} \tag{9}$$

Proof. As in Lemma 3.2, delete one edge and inspect the component not containing h . For deletions from X , these components are

$$\begin{array}{c|c} X-p & \{0, 1\} \\ X-a_i & \{2i+1\} \\ X-b_i & \{2i+2, 2i+3\} \quad (0 \leq i < m-1) \\ X-b_{m-1} & \{2m\}. \end{array}$$

For deletions from Y , they are

$$\begin{array}{c|c} Y-a_0 & \{0, 2m\} \\ Y-a_i & \{2i\} \quad (1 \leq i < m) \\ Y-c_0 & \{2m, 0, 1\} \\ Y-c_i & \{2i, 2i+1\} \quad (1 \leq i < m) \\ Y-t & \{2m\}, \end{array}$$

and for deletions from Z , they are

$$\begin{array}{c|c} Z-b_i & \{2i+1, 2i+2\} \quad (0 \leq i < m-1) \\ Z-b_{m-1} & \{2m-1, 2m, 0\} \\ Z-d_i & \{2i+1\} \\ Z-t & \{0\}. \end{array}$$

The target-tree edges crossing these cuts are exactly those in (8) and (9). $\square$

Proposition 4.3. *For every odd $n \geq 5$, the block array $\mathcal{A}_n(X, Y, Z, W)$ formed from (7) has no successful selection.*

Proof. Assume that a successful selection exists. Let Ω be the common omission multiset given by Proposition 2.2; it has size $2m$. Let $e \in Y$ and $f \in Z$ be the elements omitted by the two boundary matchings.

Every a_i has the unique target d_i on the Z boundary, and every b_i has a unique target on the Y boundary. Hence each a_i and each b_i occurs at most once in Ω . The element p occurs at most twice, because it has only two possible targets on either boundary. Let ν be the multiplicity of p .

We first determine the possible omitted boundary elements. When $\nu = 2$, the two occurrences of p use c_0, t on the Y boundary and d_0, t on the Z boundary. Consequently b_{m-1} and a_0 cannot occur in Ω , respectively. The size of Ω then forces

$$\Omega = \{p, p\} \uplus (A \setminus \{a_0\}) \uplus (B \setminus \{b_{m-1}\}),$$

and the boundary omissions are

$$e = a_0, \quad f = b_{m-1}.$$

Suppose next that $\nu = 1$. Exactly one element $q \in A \cup B$ is absent from Ω .

If $q = b_j$, then all a_i force all of D on the Z boundary. Thus p must use t , and $f = b_j$. When $j < m - 1$, the elements b_i with $i \neq j, m - 1$ force $C \setminus \{c_0, c_{j+1}\}$, the element b_{m-1} forces t , and p consequently forces c_0 . Only the choice associated with a_{j+1} remains unresolved, so

$$e \in \{a_{j+1}, c_{j+1}\}.$$

When $j = m - 1$, the remaining b_i force $C \setminus \{c_0\}$, and the choices involving p and a_0 give

$$e \in \{a_0, c_0, t\}.$$

If $q = a_j$, then all b_i force $C \setminus \{c_0\}$ together with t on the Y boundary. Hence p uses c_0 , and $e = a_j$. If $j \geq 1$, the elements a_i with $i \neq j$ force $D \setminus \{d_j\}$, and p forces t . Only b_{j-1} and b_j can use d_j , whence

$$f \in \{b_{j-1}, b_j, d_j\}.$$

For $j = 0$, the exceptional choices involving p, b_0, b_{m-1} give

$$f \in \{b_0, b_{m-1}, d_0, t\}.$$

Finally, if $\nu = 0$, then $\Omega = A \cup B$. On the Y boundary, the b_i force $C \setminus \{c_0\}$ together with t , leaving

$$e \in \{a_0, c_0\}.$$

On the Z boundary, the a_i force all of D , leaving

$$f \in \{b_{m-1}, t\}.$$

For each case, let $\mathcal{O}_Y \subseteq Y$ and $\mathcal{O}_Z \subseteq Z$ denote the sets of possible values of e and f just obtained, and define

$$\Gamma_Y(\mathcal{O}_Y) = \bigcup_{e \in \mathcal{O}_Y} E_W^Y(e), \quad \Gamma_Z(\mathcal{O}_Z) = \bigcup_{f \in \mathcal{O}_Z} E_W^Z(f).$$

Substitution into (9) gives

case	$\Gamma_Y(\mathcal{O}_Y)$	$\Gamma_Z(\mathcal{O}_Z)$
$\nu = 2$	$\{d_{m-1}\}$	$\{c_{m-1}\}$
$\nu = 1, q = b_{m-1}$	$\{c_0, d_0, d_{m-1}, t\}$	$\{c_{m-1}\}$
$\nu = 1, q = b_j, j < m - 1$	$\{c_{j+1}, d_j, d_{j+1}\}$	$\{c_j\}$
$\nu = 1, q = a_0$	$\{d_{m-1}\}$	$\{c_0, c_{m-1}, d_0, t\}$
$\nu = 1, q = a_j, 1 \leq j < m$	$\{d_{j-1}\}$	$\{c_{j-1}, c_j, d_j\}$
$\nu = 0$	$\{c_0, d_0, d_{m-1}\}$	$\{c_{m-1}, t\}$

The two completion sets in every row are disjoint. Proposition 2.2 would require the bottom-right selection to belong to both, a contradiction. $\square$

Proof of Theorem 1.2. Proposition 3.3 applies when n is even, and Proposition 4.3 applies when n is odd. $\square$

Proof of Corollary 1.3. Orient the edges of G_n arbitrarily, form its signed incidence matrix, and delete the row corresponding to the hub. The resulting matrix has n rows, $2n$ integral columns, and represents the cycle matroid of G_n over every field. Indeed, the columns indexed by a cycle satisfy the signed cycle dependence, whereas the columns indexed by a forest are independent, as follows by successively examining a non-hub leaf of a nonempty forest. This is the standard incidence representation of a graphic matroid; see, for example, [5, Chapter 5]. The four spanning trees in the appropriate parity construction therefore give four column bases, and Theorem 1.2 supplies the required array. $\square$

5. THE RANK-FOUR MEMBER

The case $m = 2$ of the even construction is the wheel with a four-cycle rim. It has the following explicit 0–1 realization over an arbitrary field $\mathbb{F}$. Let u_1, u_2, u_3, u_4 be the standard basis of $\mathbb{F}^4$, and set

$$\begin{aligned} v_1 &= u_2 + u_4 = (0, 1, 0, 1), & v_2 &= u_1 + u_3 = (1, 0, 1, 0), \\ v_3 &= u_1 + u_2 + u_4 = (1, 1, 0, 1), & v_4 &= u_1 + u_2 + u_3 = (1, 1, 1, 0). \end{aligned}$$

Define

$$\begin{aligned} X &= (u_1, u_2, u_3, u_4), & Y &= (u_4, u_3, v_1, v_2), \\ Z &= (u_2, u_1, v_3, v_4), & W &= (v_1, v_2, v_3, v_4). \end{aligned}$$

The parentheses specify an order only when the vectors are used as matrix columns. In the displayed orders, the determinants are 1, 1, 1, -1 , respectively. Thus all four sets are bases over every field, and the counterexample is $\mathcal{A}_4(X, Y, Z, W)$.

In the notation of Section 3, the four edge classes are

$$A = \{u_4, u_3\}, \quad B = \{u_1, u_2\}, \quad C = \{v_1, v_2\}, \quad D = \{v_3, v_4\}.$$

For a direct verification of the graphic representation, let

$$R = [u_1 \ v_1 \ u_2 \ v_2 \ v_3 \ u_4 \ v_4 \ u_3]$$

and

$$P = \begin{pmatrix} -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & -1 & 0 \end{pmatrix}.$$

If $e_1, \dots, e_4$ is the standard basis of $\mathbb{F}^4$, then $\det P = 1$ and

$$PR = [-e_1 \mid e_2 \mid e_3 \mid -e_4 \mid -e_1 + e_2 \mid e_2 - e_3 \mid e_3 - e_4 \mid e_1 - e_4],$$

which is a reduced signed incidence matrix of the wheel.

These four bases do not themselves contradict Rota's basis conjecture. If they are used as four column-constant bases, the following arrangement has basis rows:

	X	Y	Z	W
1	u_1	u_4	u_2	v_2
2	u_2	u_3	u_1	v_1
3	u_3	v_1	v_3	v_4
4	u_4	v_2	v_4	v_3

The determinant of each displayed row, read as a matrix of four columns, is 1. The obstruction in Theorem 1.2 therefore depends on varying the cell basis in both array directions.

6. POSITIVE RESULTS

6.1. Low rank. We begin with an elementary list-coloring lemma.

Lemma 6.1. *An even cycle is properly colorable from arbitrary two-element color lists on its vertices.*

Proof. If all lists are equal, alternate their two colors. Otherwise choose an edge uv with $L(u) \neq L(v)$, delete that edge, and assign to u a color in $L(u) \setminus L(v)$. Color greedily along the resulting path from u to v . Consecutive vertices receive different colors, and the final color at v differs from the color at u because the latter does not belong to $L(v)$. $\square$

Proposition 6.2. *Every 2×2 array of bases of a rank-two matroid has a successful selection.*

Proof. Delete the loops and replace every remaining element by its parallel class. Each cell basis becomes a two-element list of distinct classes. Join two cells when they lie in a common row or column; the four cells form a 4-cycle. Apply Lemma 6.1, and in each cell choose the unique element of its basis lying in the assigned parallel class. The two selected elements in every row and column belong to distinct parallel classes and hence form a basis. $\square$

Rank zero is vacuous and rank one is immediate. It remains to prove the disconnected rank-three case in Theorem 1.4.

Proposition 6.3. *Let M be a rank-three matroid such that $M \setminus L(M)$ is disconnected. Every 3×3 array of bases of M has a successful selection.*

Proof. Loops occur in no basis, so delete them. The positive ranks of the connected components are either $1 + 1 + 1$ or $1 + 2$.

In the first case, every cell basis contains one element from each component. Label the nine cells by the three components according to a Latin square of order three, and in each cell select the element from the indicated component. Each row and column then contains one element from each component and hence is a basis.

In the $1 + 2$ case, select the element from the rank-one component in the three cells of a permutation matrix, say the main diagonal. The six off-diagonal cells form a 6-cycle under row-column adjacency. In each off-diagonal cell, the two elements from the rank-two component determine a two-element list of parallel classes. Lemma 6.1 selects two nonparallel elements from that component in every row and column. Together with the selected element from the rank-one component, they form a basis of M . $\square$

Proof of Theorem 1.4. Combine Proposition 6.2, Proposition 6.3, and the rank-zero and rank-one observations. $\square$

Deleting loops does not change any cell basis. Moreover, a successful selection in the simplification of a matroid lifts cell by cell, because a basis contains at most one element from each parallel class. Consequently, any rank-three counterexample has a simple connected rank-three minor obtained by deleting loops and simplifying parallel classes.

6.2. Partition matroids. Let $E = E_1 \uplus \cdots \uplus E_k$, and let $r_1, \dots, r_k$ be nonnegative integers. The associated partition matroid has as its bases the sets B satisfying

$$|B \cap E_\ell| = r_\ell \quad (1 \leq \ell \leq k).$$

Its rank is $n = r_1 + \cdots + r_k$.

Proposition 6.4. *Every array of bases of a partition matroid has a successful selection.*

Proof. Identify the cells of the array with the edges of $K_{n,n}$. Decompose $K_{n,n}$ into n perfect matchings. For each ℓ , let H_ℓ be the union of r_ℓ of these matchings, using disjoint groups of matchings for different values of ℓ . Then the graphs $H_1, \dots, H_k$ partition the cells, and H_ℓ is r_ℓ -regular.

For an edge ij of H_ℓ , assign the list

$$L_{ij} = B_{ij} \cap E_\ell,$$

which has size r_ℓ . Galvin's list-edge-coloring theorem gives a proper edge coloring of H_ℓ from these lists [3]. In cell ij , select the color assigned to the corresponding edge. At every row or column vertex, the r_ℓ incident edges of H_ℓ receive distinct elements of E_ℓ . After combining all values of ℓ , each row and column contains exactly r_ℓ distinct elements of every part E_ℓ , and therefore is a basis. $\square$

Corollary 6.5. *A rank- n counterexample has at least $n + 2$ nonloop elements.*

Proof. Delete the loops. A rank- n matroid on n elements is free and hence is a partition matroid. Suppose instead that $|E| = n + 1$. Let C be the set of coloops, put $N = E \setminus C$, and write $r = n - |C|$. Then $|N| = r + 1$. For every $e \in N$, the element e is not a coloop, so some basis excludes it. Since every basis contains C , it follows that $N \setminus \{e\}$ is a basis of $M|N$. Hence

$$M = \bigoplus_{e \in C} U_{1,1} \oplus U_{r,r+1},$$

a partition matroid. Proposition 6.4 now applies. $\square$

7. COMPUTER-ASSISTED RESULTS

The results in this section depend on exhaustive finite computations. The accompanying supplement, available from <https://boonsuan.github.io/kahn-supplement.zip>, contains the C++17 and Python source files, a build-and-run script, SHA-256 checksums, and a transcript of the reference run. Running `run_all.sh` produces a fresh transcript; the recorded environment used g++ 14.2.0, Python 3.13.5, and 64-bit Linux. Every decision is made by exact set, integer, or bitset operations, with no floating-point arithmetic. The archive also contains independent finite checks of the wheel constructions in several ranks, but Theorem 1.2 is proved deductively. One additional search in $\text{PG}(2, 2)$ and $\text{PG}(2, 3)$ is exploratory and is not used in any theorem below.

7.1. Rank-three binary matroids. Let F_7 denote the Fano matroid.

Proposition 7.1 (computer-assisted). *Every 3×3 array of bases of F_7 has a successful selection.*

Exhaustive computation. The Fano matroid has 28 bases. An ordered row pattern is a triple of cell bases, so there are

$$28^3 = 21,952$$

row patterns. For a row pattern $p = (B_1, B_2, B_3)$, the program computes

$$R(p) = \{(x_1, x_2, x_3) : x_j \in B_j \text{ for all } j, \{x_1, x_2, x_3\} \text{ is a basis of } F_7\}.$$

For three row patterns p, q, r , the corresponding array has a successful selection if and only if there are

$$a \in R(p), \quad b \in R(q), \quad c \in R(r)$$

such that $\{a_j, b_j, c_j\}$ is a basis for each $j \in \{1, 2, 3\}$.

The program stores these sets as bitsets. It constructs the 168 automorphisms of F_7 from the standard binary representation. Their action, together with the six permutations of the columns, reduces the first row to 36 orbits. The search therefore examines

$$36 \cdot 21,952 = 790,272$$

first-two-row cases. For each such case, it tests all 21,952 possibilities for the third row, for a total of

$$17,348,050,944$$

bitset intersection tests. Every test is positive. $\square$

Proof of Theorem 1.5. Simplification preserves rank and binary representability. Every simple rank-three binary matroid is a restriction of F_7 , because its elements may be represented by distinct nonzero vectors of $\mathbb{F}_2^3$. Regard the nine cell bases as bases of F_7 and apply Proposition 7.1. The selected elements lie in the restriction, and independence in a restriction is inherited from F_7 . Finally, lift each selected parallel class to the unique representative in the corresponding original cell basis. $\square$

Proof of Corollary 1.6. Theorem 1.5, together with Theorem 1.4, excludes binary counterexamples of rank at most three. The rank-four example in Theorem 1.2 is graphic and hence regular and binary. Every regular or graphic matroid is binary, so rank four is minimal in each of the three classes. $\square$

7.2. Restricted minimality. The first search concerns only the four bases displayed in Section 5.

Proposition 7.2 (computer-assisted). *Every 4×4 array using at most three of the four bases X, Y, Z, W from Section 5 has a successful selection.*

Exhaustive computation. There are four three-element subsets of $\{X, Y, Z, W\}$, and arrays using one or two basis types are included in these cases. For a fixed triple, a row pattern is a word of length four over three basis types, giving $3^4 = 81$ row patterns. Row permutations preserve the existence of a successful selection, so it suffices to inspect the

$$\binom{81 + 4 - 1}{4} = 1,929,501$$

multisets of four row patterns. For every row pattern, the program enumerates all choices that form a row basis and then backtracks through the four rows while maintaining column independence. No counterexample occurs for any of the four triples. $\square$

Thus all four displayed bases are necessary within this particular four-base family. The next result concerns block arrays in the two smallest wheels.

Proposition 7.3 (computer-assisted). *For $n \in \{4, 5\}$, every block array $\mathcal{A}_n(X, Y, Z, W)$ of spanning trees of G_n that uses at most three distinct trees has a successful selection.*

Exhaustive computation. The wheels G_4 and G_5 have 45 and 121 spanning trees, respectively. Their dihedral automorphism groups have 8 and 15 orbits on the choice of X . After fixing an orbit representative for X , transposition of the block array allows the restriction $Y \leq Z$ in a fixed enumeration of the spanning trees. The program consequently checks

$$8 \binom{46}{2} 45 = 372,600 \quad \text{and} \quad 15 \binom{122}{2} 121 = 13,396,515$$

representative quadruples for G_4 and G_5 , respectively. It applies the necessary-and-sufficient criterion in Proposition 2.2. Every counterexample found uses four distinct spanning trees. $\square$

Proposition 7.4 (computer-assisted). *No rank-four graphic matroid on at most seven elements admits a counterexample of the form $\mathcal{A}_4(X, Y, Z, W)$. Consequently, the eight-edge wheel example is ground-set-minimal among rank-four graphic counterexamples of this block form.*

Exhaustive computation. It is enough to consider simple graphic matroids: loops occur in no basis, and a successful selection in a simplification lifts through the parallel classes.

A simple rank-four graphic matroid is represented by a disjoint union of connected simple graphs whose positive component ranks form one of

$$4, \quad 3 + 1, \quad 2 + 2, \quad 2 + 1 + 1, \quad 1 + 1 + 1 + 1.$$

For each part r , the program generates every labeled connected simple graph on $r + 1$ vertices and retains only direct sums with at most seven edges. This yields 731 graph models. Distinct graph models may represent the same matroid, which only enlarges the search. The program enumerates every ordered quadruple of bases X, Y, Z, W in every model, for a total of 26,342,759 quadruples.

For each quadruple, it enumerates the 20 multisets of three omissions from X and tests every compatible pair of boundary matchings and every common corner completion. Proposition 2.2 makes this test necessary and sufficient. No counterexample is found. $\square$

Proposition 7.5 (computer-assisted). *No rank-five graphic matroid on at most nine elements admits a counterexample of the form $\mathcal{A}_5(X, Y, Z, W)$. Consequently, the ten-edge wheel example is ground-set-minimal among rank-five graphic counterexamples of this block form.*

Exhaustive computation. As in Proposition 7.4, it is enough to consider simple graphic matroids. Up to vertex relabeling, the program generates every connected simple graph of ranks one through five needed to form direct sums of total rank five and at most nine edges. The possible component-rank partitions are

$$5, \quad 4 + 1, \quad 3 + 2, \quad 3 + 1 + 1, \quad 2 + 2 + 1, \quad 2 + 1 + 1 + 1, \quad 1 + 1 + 1 + 1 + 1.$$

This gives 123 graph models.

For each model and each basis X , the program applies Proposition 2.2 simultaneously to all choices of Y, Z, W . Transposition permits Y and Z to be treated as an unordered pair. For fixed X, Y, Z , the program forms the set of ground-set elements that can complete both boundary matchings; by the block criterion, $\mathcal{A}_5(X, Y, Z, W)$ is successful exactly when W meets this set. In total, the search examines 2,401,139 choices of X and the unordered pair $\{Y, Z\}$. No counterexample is found. $\square$

8. QUESTIONS LEFT OPEN BY THE PRESENT WORK

The results above leave several natural questions.

- (1) Does every 3×3 array of bases of an arbitrary rank-three matroid have a successful selection?
- (2) Can a counterexample use only three distinct cell bases? Propositions 7.2 and 7.3 rule this out only in the stated restricted settings.

- (3) Is $2n$ the minimum ground-set size of a rank- n graphic block counterexample? Propositions 7.4 and 7.5 answer this for $n = 4, 5$. What lower bounds hold without the graphic or block-form restrictions?
- (4) Can the parity-dependent wheel constructions be subsumed by a single graphic or matroidal obstruction theorem?
- (5) Which quadruples of spanning trees of a wheel yield block counterexamples under Proposition 2.2?

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